DRL-Based Weakly Supervised Traffic Anomaly Detection for IoT Networks

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***Abstract*—With the proliferation of Internet of Things (IoT) devices, IoT systems face increasingly severe threats from large-scale cyberattacks during network communications. To effectively recognize potential network threats and ensure the security of IoT, network traffic anomaly detection has been widely studied. Due to the high cost of manual labeling in real-world scenarios, only limited network traffic data is explicitly labeled as abnormal or normal. However, most existing anomaly detection methods struggle to enhance detection accuracy with minimal supervision and perform ineffectively at identifying potential unknown anomalies in unlabeled data. To address these limitations, this paper proposes WADE, a *Weakly supervised Anomaly DEtection* method based on Deep Reinforcement Learning (DRL). WADE enables simultaneous identification of known and unknown anomaly traffic with limited labeled data. Specifically, it incorporates a Dueling Q-network architecture and introduces a novel reward optimization mechanism that: 1) strengthens feature extraction from unlabeled data and 2) elevates decision accuracy via dynamic reward adaptation. Extensive experiments on IoT traffic datasets demonstrate that WADE outperforms four peer weakly supervised methods, achieving performance improvements of 12.2% in Area Under the Precision-Recall Curve (AUC-PR) and 6.6% in Area Under the Receiver Operating Characteristic Curve (AUC-ROC), which validate the effectiveness of WADE in safeguarding IoT systems.**

***Index Terms*—Internet of Things (IoT), network traffic, anomaly detection, deep reinforcement learning, weakly supervised learning.**

I. Introduction

With the rapid advancement of network technologies and widespread adoption of connected devices, Internet of Things (IoT) has been increasingly integrated into diverse domains, such as smart city, healthcare and manufacturing, demonstrating immense market potential. By 2025, global IoT device count is projected to reach 75 billion [1], while IoT market size is anticipated to attain \$76.97 billion. In IoT networks, the deployment of large-scale devices and the complexity of network communications lead to generate massive data traffic. Consequently, the risk of cyber-attacks, such as denial-of-service, on IoT systems is escalating, posing a growing threat to IoT security [2].

Although security solutions such as encryption algorithms, authentication protocols and firewalls can mitigate threats to some extent, IoT networks remain vulnerable to numerous malicious activities. Anomaly detection (AD) is a critical field in data mining that aims to identify deviant samples within a general data distribution [3]. Within IoT network scenarios, traffic anomaly detection facilitates the recognition of anomalous behaviors and potential threats, thereby enhancing the security and stability of networks [4].

In real-world scenarios, network traffic data often exhibits a static nature and encompasses multiple characteristics (features). Given the large-scale deployment and inherent resource limitations of IoT devices, only limited traffic data is typically labeled as normal or abnormal. These labeled abnormal data with specific classes (types) is regarded as known anomalies. On the other hand, extensive unlabeled data encompasses both known classes of anomalies and previously unseen classes of anomalies, referred as unknown anomalies [5]. Recently, Deep Learning (DL) algorithms have emerged as pivotal tools in cybersecurity applications, offering advanced solutions for network traffic anomaly detection. DL can automatically extract data features without extensive and laborious feature selection [6], which is well-suited for handling high-dimensional and large-scale network traffic. Therefore, we focus on DL-based anomaly detection to identify known and unknown anomaly traffic, as illustrated in Fig. 1.

Considering the anomaly detection task involving large-scale unlabeled data (consisting mostly of normal data) and limited

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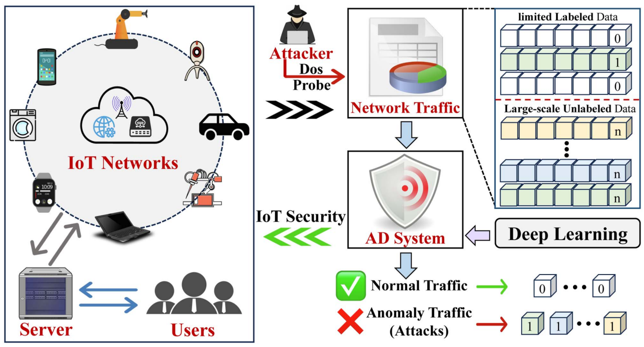

Fig. 1. An illustration of IoT network anomaly detection, where *0* and *1* respectively represent labels for normal and abnormal traffic, *n* represents no labels. The network attacks generate abnormal traffic affecting network security. Through DL empowered AD system, both known and unknown anomalies are identified to safeguard the security of network communication.

labeled anomalies that only cover partial classes of anomalies, it is consequently hard to employ supervised learning methods like [7], [8], [9] that rely on comprehensive labeled data for training anomaly detection models. Unsupervised learning methods like [10], [11], [12] train anomaly detection models without using labeled data, which can detect diverse anomalies (different classes of anomalies) since they are not constrained by labeled data [13]. However, the detection accuracy of these methods decreases as the false positive rate increases, rendering them inadequate to meet specific requirements. Semi-/weakly supervised learning methods utilize only limited labeled anomalies and can provide valuable information for training [14], which are thus more relevant to this study. These methods offer significant advantages in reducing computational costs and enhancing efficiency, demonstrating the scalability for large-scale datasets. However, current semi-supervised learning methods still have certain problems. Typically, methods in [15], [16] are primarily designed to detect only known anomalies, failing to identify unknown anomalies that are absent from the training set. In contrast, methods in [3], [17], [18] can detect both known and unknown anomalies. However, during training phase, they neglect to explore potential anomalies in unlabeled data, resulting the decrease in accuracy of unknown anomaly detection.

To address the aforementioned issues of semi-supervised methods based on deep learning, DPLAN [13] leverages the Deep Reinforcement Learning (DRL) framework to learn anomalies in both labeled and unlabeled data. DRL autonomously learns normal and abnormal traffic patterns from massive data without extensive manual annotation, making it ideal for large-scale network traffic analysis. Moreover, DRL adapts to dynamic network environments through real-time decision-making, improving detection accuracy for both known and unknown anomalies. However, the Deep Q-Network (DQN) used in the agent of DPLAN struggles to effectively model state-instance relationships. For network traffic anomaly detection, different traffic states have varying importance in identifying anomalies, while DQN's direct learning of Q-values fails to separate state values from action advantages, potentially ignoring critical state features. Additionally, DQN struggles in high-dimensional state spaces, limiting its ability to extract action-relevant information from complex states. Furthermore, the combined reward function in DPLAN cannot fully explore unknown anomalies in unlabeled data, for which the reward mechanism fails to provide optimal training rewards.

Motivated by these challenges and inspired by [13], this paper further proposes a DRL-based weakly supervised anomaly detection method (WADE) to fully explore unlabeled data and identify unknown anomaly traffic for IoT networks. Firstly, Dueling Q-network (Dueling DQN) is adopted [19] as network architecture of the agent, enhancing its modeling capacity and learning efficiency by accurately capturing relationships between states and actions under anomaly conditions. Secondly, a novel reward optimization mechanism is designed to enhance learning from unlabeled data. In this mechanism, Variational Auto-Encoder (VAE) [20] is adopted to design intrinsic reward function, which improves detection performance for unlabeled data without supervision. Then an external reward function that balances internal factors (from unlabeled data) and external factors (from labeled data) is designed, which encourages the agent to learn known anomalies while amplifying rewards for potential correct detections in unlabeled data. Furthermore, we employ Magnify Saltatory Reward (MSR) algorithm [21] for anomaly detection, and thus design a MSR reward optimizer that continuously monitors changes in rewards. The optimizer dynamically adjusts current reward by magnifying or decreasing it under real-time conditions.

The contributions of this paper are summarized as follows:

- A Dueling DQN-based anomaly detection framework is proposed to identify both known and unknown abnormal network traffic patterns in partially labeled data.
- A reward optimization mechanism that fully explores unlabeled data is designed, which consists of VAE-based intrinsic reward function, external reward function and MSR reward optimizer.
- Extensive experiments on real-world IoT traffic datasets demonstrate that WADE outperforms four weakly supervised methods, achieving average improvements of 12.2% in AUC-PR and 6.6% in AUC-ROC.

The structure of this paper is as follows: Section II introduces the research background and comprehensively reviews various DL-based traffic anomaly detection methods. Section III describes the tailed DRL for traffic anomaly detection task. Section IV presents the details of WADE. Section V presents experimental results. Section VI concludes this paper.

II. Research Background & Related Work

A. IoT Network Traffic Data

IoT network traffic data refers to the packets, protocol interactions and connection behaviors generated by IoT devices during network communication. The accidental operational patterns of devices make anomalous traffic indicating potential security threats. This characteristic has established IoT traffic data as a foundation for cybersecurity research [22]. Table I provides

TABLE I
IoT Network Traffic Datasets and Attack Types

Dataset	Attack Types
UNSW-NB15 [23] combines real network traffic with simulated attacks, offering 49 features including flow statistics and protocol attributes, covering modern attacks.	Analysis, Backdoor, DoS, Exploits, Reconnaissance, Fuzzers, Generic, Shellcode, Worms
KDDCUP99 [24] contains 4.9 million network connection records with 41 features.	DoS, R2l, Probe, U2r
NSL-KDD [25] improves on KDDCUP99 by eliminating the redundant data and rebalancing the sample distribution.	DoS, R2l, Probe, U2r
BoT-IoT [26] features protocol headers and communication timing, highlighting botnet communication and data leakage patterns.	DoS, Information Theft, Probe, Reconnaissance
CICIDS2017 [27] captures real network behaviors through simulation with 80+ features like packet length and protocol flags.	Brute Force, Botnet, DoS, DDoS, Heartbleed, Infiltration, Web Attacks

detailed descriptions of representative IoT network traffic datasets with their attack types.

B. Network Traffic Anomaly Detection Workflow

1) Data Preprocessing: Network traffic data contains both numerical and categorical features, where the preprocessing is required to support AD task. For categorical features, one-hot encoding is typically employed to convert them into numerical form by creating binary columns. For numerical features, Min-Max/Z-Score normalization is required to eliminate scale differences. The former scales features to [0,1] range, which is effective for bounded distributions with limited anomalies. The latter standardizes features to achieve zero mean and unit variance, exhibiting enhanced robustness against anomalies.

2) Anomaly Detection: During training, anomaly detection models learn to distinguish feature patterns between normal and anomalous traffic by optimizing network parameters. Then, anomaly scores are calculated to evaluate the anomaly degrees. In practical applications, the trained models execute real-time analysis of incoming network traffic to identify potential anomalies [28].

3) Performance Evaluation: Multiple metrics are required for comprehensive evaluation, mainly comprising:

- *Accuracy:* quantifies the ratio of correctly predicted instances, representing the overall classification efficiency.
- *Precision:* evaluates the capability to correctly identify true positive samples while considering false alarm rates.
- *Recall:* assesses the ability to detect all positive instances, reflecting the effectiveness in identifying anomalies.
- *F1-score:* provides a harmonic mean of precision and recall, offering balanced performance evaluation particularly for attack detection scenarios.
- *AUC-ROC:* assesses the model discrimination capability across all possible thresholds using the area under the receiver operating characteristic curve.
- *AUC-PR:* assesses performance via the area under the precision-recall curve, particularly valuable for imbalanced datasets where anomalies are rare.

C. Anomaly Detection Model Architecture

Considering different model architectures, network traffic anomaly detection studies are categorized into forecasting-based, reconstruction-based and hybrid approaches, summarized in Table II.

1) Forecasting-Based Models: These methods train predictive models that comprehensively capture traffic patterns. Researchers have employed Convolutional Neural Networks (CNN), Recurrent Neural Network (RNN) and Long Short-Term Memory (LSTM)-based temporal models for traffic anomaly detection. Bakhsh et al. [29] integrate Feedforward Neural Networks (FFNN), LSTM and Random Neural Networks (RandNN), where FFNN handles complex traffic patterns, LSTM captures long-term dependencies and RandNN provides dynamic adaptability. Wang et al. [30] propose an attention-weighted model combining bidirectional LSTM and self-attention mechanism, which captures packet transmission relationships by constructing a perceptual node collection graph. Moreover, multiple enhancement strategies of traditional LSTM are recently studied. Zou et al. [31] design a feature-attended multi-flow LSTM that employs an innovative feature grouping algorithm. Wang et al. [32] propose a spatial-temporal knowledge distillation framework that integrates spatial-temporal residual networks and focal loss function, effectively addressing model complexity and class imbalance issues by stacking CNN and LSTM modules.

The Graph Neural Networks (GNN)-based methods also attracted great attention recently. Caville et al. [33] leverage edge features and graph topological structure in self-supervised process for GNN-based anomaly detection. Duan et al. [34] propose the dynamic line GNN using semi-supervised paradigm, significantly enhancing performance through spatiotemporal graph modeling and message aggregation. Furthermore, to achieve lightweight AD for IoT network, Zhou et al. [35] design a graph representation learning model combining GNN and knowledge distillation. Based on Graph Attention Network (GAT) and MLP, it extracts structural and traffic features from graph and traffic data respectively, improving knowledge transfer efficiency and classification accuracy.

2) Reconstruction-Based Models: These methods detect anomalies by comparing the inputs with their reconstructed outputs. Autoencoder (AE) and its improved variants show remarkable efficacy in unsupervised anomaly detection by learning normal data distributions through encoding-decoding, while producing significant reconstruction errors for anomalous inputs. Lanvin et al. [36] introduce a novel p-value statistical analysis of reconstruction errors, enabling both accurate anomaly detection and interpretable feature localization. Then, to address the challenge of threshold setting in AE, Tang et al. [37] design an adaptive reconstruction error threshold algorithm that dynamically optimizes thresholds by jointly modeling training data distribution and anomaly priors. In addition, Zhang et al. [38] propose a Linear Autoencoder (LAE) by directly removing the activation function from AE, and introduces adversarial methods from image domain into network intrusion detection. Alrayes et al. [39] emphasis on the effectiveness of the Denoising Autoencoder (DAE) for identifying and mitigating intrusions.

TABLE II
A COMPARISON OF TRAFFIC AD METHODS WITH DIFFERENT MODEL ARCHITECTURES

Ref.	Year	Classification	Technology	Dataset	Metric	Insight
[29]	2023	Forecast.	LSTM	CIC-IOT22	ACC	Integrates FFNN, LSTM and RandNN for anomaly detection.
[30]	2023	Forecast.	LSTM	ToN-IoT	ACC	Attention-weighted model that captures packet transmission relationships.
[31]	2022	Forecast.	LSTM	BoT-IoT	ACC	Designs a feature-attended multi-flow LSTM.
[32]	2024	Forecast.	CNN, LSTM	CICIDS2017	F1	Integrates spatial-temporal residual networks and focal loss.
[33]	2022	Forecast.	GNN	UNSW-NB15	F1	Leverages edge features and graph structure in self-supervised process.
[34]	2022	Forecast.	GNN	NIDS	F1	Proposes a dynamic line GNN that employs semi-supervised learning.
[35]	2024	Forecast.	GNN, GAT	CICIDS2017	F1	Leverages GAT and MLP to extract structural and traffic features.
[36]	2023	Recon.	AE	CICIDS2017	ACC	Introduces a novel p-value analysis of reconstruction errors.
[37]	2024	Recon.	AE	NSL-KDD	ACC	Designs an adaptive reconstruction error threshold algorithm.
[38]	2024	Recon.	LAE	NSL-KDD	F1	Introduces adversarial methods from image domain into intrusion detection.
[39]	2024	Recon.	DAE	NSL-KDD	ACC	Leverages DAE for IoT intrusion detection.
[40]	2024	Recon.	GAN	UNSW-NB15	F1	Employs conditional GAN to generate minority class samples.
[41]	2024	Recon.	GAN	Kaggle	F1	Uses PCA for feature selection and GAN for generating synthetic samples.
[42]	2025	Recon.	GAN	NSL-KDD	F1	Combines advantages of denoising autoencoders and wasserstein GAN.
[43]	2023	Hybrid	CNN, BERT	ICPS	F1	Combines 1D-CNN and BERT to extract complex payload dependencies.
[44]	2024	Hybrid	CNN, GA	Edge-IIoT	ACC	Transforms network data into image formats for feature analysis.
[45]	2024	Hybrid	CNN, DNN	UNSW-NB15	F1	Base layer comprises CNN, LSTM and GRU, top layer employs DNN.
[46]	2024	Hybrid	AE, DNN	CICIDS2018	ACC	Employs autoencoder variants for unsupervised feature extraction.
[47]	2024	Hybrid	GAN, DNN	NSL-KDD	ACC	Constructs a classification model by integrating GAN and deep KNN.

Generative Adversarial Networks (GAN) and its variants have shown remarkable effectiveness in addressing data imbalance issues. Wang et al. [40] employ conditional tabular GAN to generate minority class samples, significantly improving multiclass detection accuracy through enhanced data diversity and particularly excelling in detecting rare attack types. Changala et al. [41] integrate Principal Component Analysis (PCA) for feature selection, enabling dimensionality reduction while extracting fundamental traffic patterns. It then leverages GAN to learn the distribution of normal traffic and generate synthetic samples for anomaly detection. AE-WGAN [42] creatively combines advantages of denoising autoencoders and wasserstein GAN, achieving superior performance.

3) Hybrid Models: These methods integrate multiple DL paradigms to significantly improve model performance and adaptability. Yang et al. [43] innovatively combine 1D-CNN and BERT, effectively addressing the challenge of extracting complex payload dependencies. Latif et al. [44] transform network data into image formats for CNN-driven feature analysis, employs GA for hyperparameter optimization and ultimately utilizes ensemble techniques. Similarly, ENIDS [45] adopts two-layer ensemble architecture, where the base layer comprises CNN, LSTM and Gated Recurrent Unit (GRU) as base learners, the top layer employs DNN for meta-learning. Furthermore, Kunang et al. [46] utilize multiple autoencoder variants for unsupervised feature extraction, achieving near 100% ACC. Zhong et al. [47] design DNN and contrastive learning-based framework integrating GAN and deep K-Nearest Neighbor (KNN), which can implicitly defend against adversarial attacks and detect fine-grained unknown attacks.

D. Anomaly Detection Learning Strategy

This section reviews different anomaly detection learning strategies including continual learning, online learning and DRL for dynamic IoT environments, summarized in Table III.

1) Continual Learning (CL): For dynamic IoT environments, CL-based methods enable anomaly detection systems to continuously evolve through progressive knowledge accumulation and adaptation to emerging threats. Memory replay represents a crucial direction in CL studies. Channappayya et al. [48] propose the augmented memory replay framework innovatively integrating class-balanced reservoir sampling (CBRS) and perturbation-assisted parameter approximation (PAPA). CBRS addresses class imbalance issues, while PAPA reduces training time and enhances practical deployment. For privacy preserving, SPIDER [49] combines gradient projection memory with semi-supervised continual learning, attaining performance comparable to fully supervised approaches using merely 20% labeled data.

To address the concept drift issue in IoT networks, Amalapuram et al. [50] conduct a comparison of elastic weight consolidation and gradient episodic memory across various learning environments. Experimental results reveal that a hybrid strategy enhanced by experience forgetting-aware memory filling, achieves optimal performance in domain incremental learning scenarios. Zhang et al. [51] propose the strategic CL that incorporates a dynamic memory buffer and strategic forgetting mechanism. By selecting representative drifting samples and eliminating outdated knowledge, it enables continuous adaptation to novel network threats.

TABLE III
A Comparison of Traffic AD Methods With Different Learning Strategies

Ref.	Year	Classification	Dataset	Metric	Insight
[48]	2023	CL	NSL-KDD	AUC	Integrates class-balanced sampling and perturbation-assisted parameter approximation.
[49]	2024	CL	CICIDS2017	AUC	Combines gradient projection memory with semi-supervised continual learning.
[50]	2022	CL	KDDCUP99	F1	Incorporates experience forgetting-aware memory filling technology.
[51]	2024	CL	NSL-KDD	F1	Incorporates a dynamic memory buffer and strategic forgetting mechanism.
[52]	2024	CL	CICIDS2018	Recall	A federated learning-based method to mitigate catastrophic forgetting problem.
[53]	2025	CL	CICIDS2017	F1	Leverages learning-based feature extractor and PCA reconstruction to identify attacks.
[54]	2021	OL	UNSW-NB15	FPR	Combines the online and offline learning paradigms.
[55]	2022	OL	DoS	F1	Dynamically adjusts the structure of online DNN using hedge weighting mechanism.
[56]	2024	OL	BoT-IoT	F1	A distributed self-supervised federated intrusion detection algorithm with DRNN.
[28]	2024	OL	NSL-KDD	F1	Integrates AE-based feature extraction with a customized contrastive loss function.
[57]	2024	OL	BoT-IoT	AUC	A fully online self-supervised learning architecture that makes autonomous decisions.
[58]	2025	OL	NSL-KDD	F1	A novel drift-adaptive online ensemble learning framework.
[59]	2021	DRL	NSL-KDD	F1	Integrates autoencoders with double deep Q-networks.
[60]	2023	DRL	ToN-IoT	ACC	Applies conditional VAE enhanced DQN for reconstructing traffic features.
[61]	2024	DRL	UNSW-NB15	F1	Comprises an traffic processing module and a DQN-based heuristic learning network.
[62]	2023	DRL	NSL-KDD	Recall	Integrates conditional tabular GAN with multi-agent reinforcement learning.
[63]	2024	DRL	NSL-KDD	AUC	The distributed multi-agent reinforcement learning framework.
[64]	2024	DRL	NSL-KDD	ACC	Proposes to train DRL models using synthetic data generated by GANs.
[65]	2024	DRL	UNSW-NB15	F1	Introduces a universal state representation and adaptive reward mechanism.

Jin et al. [52] propose a federated learning-based intrusion detection framework, utilizing class gradient balance loss to reduce the learning of new classes and sample label smoothing loss to enhance the model memory for old classes. It then employ relay client fusing sample reconstruction to mitigate the spread of catastrophic forgetting globally without compromising data privacy. Latest research [53] leverages a learning-based feature extractor that dynamically updates representations and a novelty detector leveraging PCA reconstruction mechanism to identify emerging attacks, achieving efficient continual learning for streaming data.

2) Online Learning (OL): Online learning contributes to real-time anomaly detection in dynamic environments through model adaptation. Odiathevar et al. [54] propose a offline-online framework where the offline model establishes benchmark features of normal traffic and online component dynamically adjusts detection threshold. Then, Wahab et al. [55] propose an integrated detection-adaptation framework that employs PCA-based feature variance analysis for concept drift detection. The system dynamically optimizes its online DNN architecture through a hedge weighting mechanism, reducing in false positives while maintaining stable long-term detection accuracy. Gelenbe et al. [56] propose a novel distributed intrusion detection algorithm that features online self-supervised federated learning capabilities and employs the Dense Random Neural Networks (DRNN). Moreover, Zhang et al. [28] propose an online intrusion detection system that integrats the pseudo-label autoencoder with customized contrastive loss function, effectively achieving autonomous evolution of features and novel attacks detection. Nakip et al. [57] present an online self-supervised framework that leverages deep stochastic neural networks to enable autonomous learning, completely eliminating dependency on pre-training and manual annotation. It demonstrates potential for real-time monitoring in resource-limited scenarios. Latest research [58] introduces a novel drift-adaptive ensemble learning framework comprising three stages (preprocessing, base model learning and online ensemble), specifically combining four online learning methods and optimizing key parameters through particle swarm optimization.

3) Deep Reinforcement Learning: DRL has demonstrated human-level capability, making it increasingly important for anomaly detection [72]. In term of DQN and its variants, Dong et al. [59] develop a semi-supervised framework combining autoencoder with double DQN. Specifically, the current network is responsible for feature reconstruction and classification, while the target network leverages K-Means clustering for auxiliary predictions. DC-IDS [60] further focuses on two sub-tasks (known traffic fine-grained classification and unknown attacks identification) and applies conditional VAE-enhanced DQN for reconstructing traffic features. Furthermore, Shen et al. [61] propose a zero-day attacks detection system for social IoT, which first processes the traffic samples and computes similarity scores. Then the LSTM-based DQN is adopted to optimize the heuristic learning network strategy by maximizing cumulative rewards.

For other structures, Mouyart et al. [62] integrate conditional tabular GAN with multi-agent reinforcement learning, utilizing generated synthetic samples to balance the dataset. Similarly, Louati et al. also [63] employ a distributed multi-agent reinforcement learning framework, which harnesses collaborative inter-agent communication, parallelized data

TABLE IV
A COMPARISON OF TABULAR AD METHODS WITH DIFFERENT LEARNING PARADIGMS

Ref.	Year	Classification	Dataset	Metric	Insight
[10]	2022	Unsupervised	Tabular	F1	Couples two loss functions to extract learning signals from normal and anomalous data.
[11]	2023	Unsupervised	UNSW-NB15	AUC	Leverages scale-based labeled data to train neural networks.
[66]	2023	Unsupervised	Tabular	AUC	First studies zero-shot AD tasks for tabular, introducing the adaptive central representation.
[67]	2024	Unsupervised	Tabular	AUC	Integrates semantic information from column names to generate augmented views.
[68]	2024	Unsupervised	NSL-KDD	AUC	Incorporates learnable masking strategy to determine input-specific masking patterns.
[69]	2025	Unsupervised	Backdoor	AUC	Proposes decomposed representation learning and remaps data into a constrained space.
[70]	2025	Unsupervised	Tabular	AUC	Leverages LLMs and transforms tabular data into standardized text format.
[13]	2021	Semi-supervised	UNSW-NB15	AUC	Employs DQN to detect both known and unknown anomalies.
[14]	2022	Semi-supervised	UNSW-NB15	AUC	Detects both discrete anomalies and partially identified group anomalies.
[18]	2022	Semi-supervised	NSL-KDD	AUC	Constructs representations to enhance scores learning.
[3]	2023	Semi-supervised	Tabular	AUC	Determines the anomaly threshold by estimating the score distribution.
[17]	2023	Semi-supervised	KDDCUP99	AUC	Employs relation neural network to minimize three-way prediction loss.
[71]	2024	Semi-supervised	Tabular	AUC	Generates pseudo-anomalies using few labeled anomalies and large-scale unlabeled data.

preprocessing and cloud-based computational resources. Then, Strickland et al. [64] propose to train models using synthetic data generated by GAN. Experimental results demonstrate that DRL-based models achieve superior detection performance of minority classes when trained on GAN-generated data rather than imbalanced real-world datasets. Furthermore, He et al. [65] introduce a universal state representation and adaptive reward mechanism, achieving cross-dataset generalization. The system converts raw traffic into standardized environmental states, ensuring dimensional independence.

E. Tabular Anomaly Detection

Given the structural similarities between tabular and network traffic data, AD techniques for tabular can be effectively adapted for network traffic data. Therefore, this section reviews the advanced tabular AD methods, as summarized in Table IV.

For unsupervised learning paradigm, Qiu et al. [10] design two coupled loss functions to extract learning signals from noisy datasets and utilize iterative block coordinate updates to optimize model parameters and label inferences, demonstrating strong generalization capabilities across tabular, image and video AD tasks. SLAD [11] utilizes scale-annotated data to train models and introduces a data-driven supervision mechanism to derive anomaly scores directly from loss values. To overcome data augmentation challenges for contrastive learning-based tabular AD, Tao et al. [67] leverages semantic information from column names to generate effective augmented views. Yin et al. [68] adapt masked modeling paradigms from vision/language domains to tabular data and propose MCM, which incorporates learnable masking strategy to determine input-specific masking patterns and employs ensemble learning with the diversity loss to capture multiple correlations in training data. Ye et al. [69] address the data entanglement issue in reconstruction-based methods by introducing decomposed representation learning. It remaps data into a designed constrained space to capture latent common normal patterns and identify anomalous patterns. Kloft et al. [66] introduce the adaptive central representation, which adapts to a set of interrelated training data distributions, enabling automatic zero-shot generalization to unseen anomaly detection tasks. Furthermore, Tsai et al. [70] leverage large language models (LLMs) for unsupervised tabular anomaly detection. By converting tabular data into textual formats, fine-tuned LLMs process serialized input and calculate anomaly scores based on negative log-likelihood.

For semi-/weakly supervised learning paradigm, DevNet [73], PReNet [17], EDAE [3] DPLAN [13], FEAWAD [18] and Dual-MGAN [14] are latest methods. DevNet assumes a gaussian prior on anomaly scores and generates reference score samples accordingly. The deviation loss ensures normal data clusters around reference scores while maintaining significant separation from labeled anomaly scores. PReNet extends DevNet by employing relation neural network to minimize the three-way prediction loss. It learns both pairwise relation features and anomaly scores of randomly selected instance pairs. FEAWAD improves on DevNet by integrating AE to encode inputs. It uses hidden representation, reconstruction residual vector and reconstruction errors to enhance scores learning. EDAE determines the anomaly threshold by estimating score distribution and employs an empirical distribution-based batch sampling strategy to enhance robustness against data contamination. Dual-MGAN detects both discrete anomalies and partially identified group anomalies. It utilizes multiple generative adversarial active learning to construct reference distribution and multiple generative adversarial over-sampling to balance minority classes. DPLAN employs DQN to detect both known and unknown anomalies by interactively exploiting labeled anomalies and actively exploring unlabeled data for rare anomalies. Additionally, Dong et al. [71] propose nearest neighbor gaussian mix-up for data augmentation, which generates additional pseudo-anomaly samples, facilitating the detection of novel anomalies.

F. Research Limitations

Despite the effectiveness of the above methods, several research limitations remain: 1) Some forecasting-based models

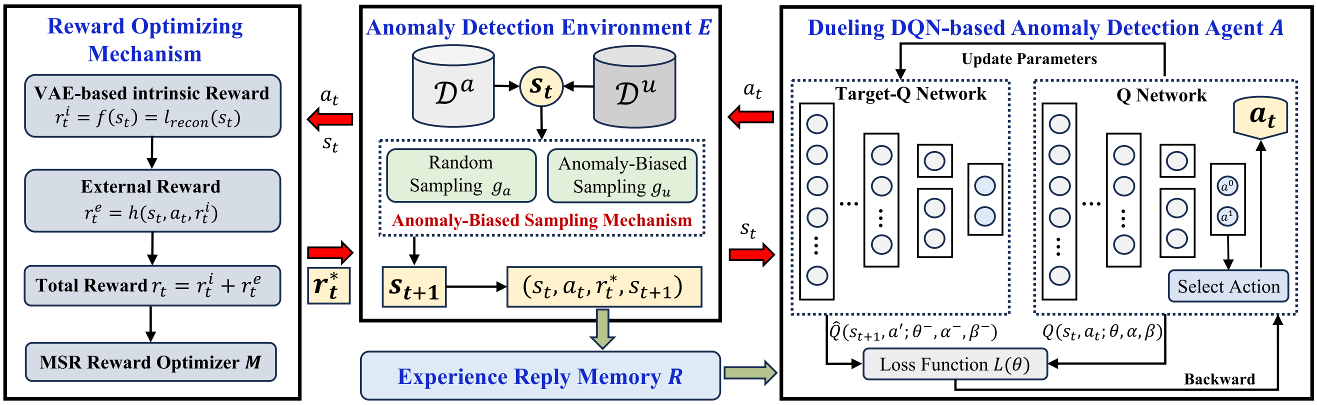

Fig. 2. Overview of DRL-based anomaly detection, which describes the process of interaction between the agent and the environment.

struggle to accurately capture rapidly changing traffic patterns, while reconstruction-based models face challenges in threshold optimization and training stability. Hybrid models are hindered in practical applications by their complex structure and high computational costs. 2) Continual learning and online learning methods struggle to maintain adaptability against large-scale traffic and complex cyber threats. DQN-based methods still need improvements in state-action estimation and complex feature extraction. 3) Unsupervised methods may suffer significant performance degradation when processing high-noise and high-dimensional traffic. Meanwhile, semi-supervised approaches show limited effectiveness in detecting unknown attacks due to inadequate exploration in unlabeled anomalies.

Therefore, we proposes a DRL-based weakly supervised network traffic anomaly detection method that adopts Dueling DQN to enhance the state-action value estimation and feature extraction, while designing novel reward mechanism to fully explore potential anomalies in unlabeled data.

III. DRL Tailored for Anomaly Detection

A. Problem Formulation

IoT network traffic data $\mathcal{D}$ is characterized by a N-dimensional feature space. During training, $\mathcal{D}$ comprises large-scale unlabeled dataset $\mathcal{D}^u$ and limited labeled known anomaly dataset $\mathcal{D}^a$, where $\mathcal{D}^u$ encompasses unknown normal data and few unknown anomalies. This study aims to train a anomaly scoring function $\varphi : \mathcal{D} \to \mathbb{R}$ that evaluates anomaly degrees of data instances including both known and unknown anomalies, where higher anomaly scores indicate higher anomaly degrees.

B. DRL-Based Anomaly Detection Framework

To identify both known and unknown attacks, we propose a DRL framework, comprising the environment, agent, action, state and reward specifically designed for anomaly detection. Based on the problem formulation, the core objective is to design the anomaly detection agent capable of effectively exploiting $\mathcal{D}^a$ to learn known anomalies, while actively exploring potential unlabeled known/unknown anomalies in $\mathcal{D}^u$.

Fig. 2 illustrates the DRL-based anomaly detection approach, where the environment E contains anomaly-biased sampling function g, external reward function h, intrinsic reward function f and MSR reward optimizer M. The agent A is driven to automatically interact with E through the optimization reward r_t^*. To this end, we transform anomaly detection task into a DRL problem by defining the following key components. For reference, Table V summarizes the symbols used in associated mathematical descriptions.

- *State space (s):* The state space s for a network traffic instance in $\mathcal{D}$ at time step t contains all features (excluding label-related attributes) from the instance, including both categorical and numerical features that covers information such as bytes per IP, session time per IP, service type, etc. Formally, the state is represented as $s = \{v^1, v^2, \ldots, v^N\}$ where N denotes the number of features in traffic instance and v^i corresponds to the value of the i-th feature.
- *Action space (a):* The action a is the decision or choice made by the agent in response to the current state s of the environment. For network traffic anomaly detection tasks, the action space $a = \{a^0, a^1\}$ represents the binary classification decision for a given state s, where a^0 corresponds to a normal classification and a^1 indicates an anomalous classification.
- *Agent (A):* The agent A is the decision-making entity that aims to learn an optimal policy for mapping states to actions by maximizing the cumulative rewards. For network traffic anomaly detection, A is implemented by a neural network to determine an optimal action out of $\{a^0, a^1\}$ for the current traffic state s. During training, it updates the network parameters to refine its policy selection. The trained network is ultimately used to evaluate the anomaly scores of the testing set.
- *Simulation Environment (E):* The environment E in anomaly detection tasks serves to emulate real-world network traffic data $\mathcal{D}$ and interacts automatically with the agent A. It contains the sampling function g and the reward function r.
- *Sampling Function (g):* The sampling function g determines the next state s_{t+1} from $\mathcal{D}$ based on the current state

TABLE V
MATHEMATICAL SYMBOLS AND DESCRIPTIONS

Symbols	Descriptions
$\mathcal{D}^a$	Labeled anomaly dataset.
$\mathcal{D}^u$	Unlabeled dataset.
N	Feature dimension of data instance.
A	Anomaly detection agent.
E	Simulation environment.
R	Reply memory.
g	Anomaly-biased sampling function.
f	Intrinsic reward function.
h	External reward function.
M	MSR reward optimizer.
s_t	Current state at time step t.
a_t	The taken action for current s_t at time step t.
s_{t+1}	Next sampled state.
r_t^i	Intrinsic reward.
r_t^e	External reward.
r_t^*	Final optimized reward.
$(s, a, r, s^{'})$	The random mini-batch sampled from R.
$a^{'}$	The taken action for the batch of next state $s^{'}$.
$Q^*(s, a)$	Q-value for the state-action pair s and a.
j	Current episode of $n\_episodes$.
$Q(s, a; \theta_j, \alpha_j, \beta_j)$	The current Q-value predicted by the main Q-network with the network parameters θ_j, α_j and β_j.
$\hat{Q}(s^{'}, a^{'}; \theta_j^-, \alpha_j^-, \beta_j^-)$	The target Q-value predicted by the target Q-network with the network parameters θ_j^-, α_j^- and β_j^-.
$\mathcal{L}_j(\theta_j, \alpha_j, \beta_j)$	Loss function for training Dueling DQN at episode j.
$Q(s, a; \theta^*, \alpha^*, \beta^*)$	The trained main Q-network with the optimal network parameters θ^*, α^* and β^*.

s_t and executed action a_t. In network traffic anomaly detection, g is designed to bias towards anomalies to enhance the learning for abnormal patterns and better detect known and unknown anomalies.

- *Reward Function (r):* For anomaly detection, the reward function r provides crucial feedback (positive/negative reward) based on the agent's correct/incorrect classification of s. This feedback mechanism in return drives A to optimize network parameters and decision-making, ultimately improving its discrimination capability between benign and malicious traffic data. Specifically, the intrinsic reward function f drives A to simultaneously learn known/unknown anomalies in $\mathcal{D}^u$, where the positive reward is provided when discovering novel anomaly state s. The external reward h encourages A to automatically learn labeled anomalies from $\mathcal{D}^a$, while amplifying learning for $\mathcal{D}^u$ based on r^i. h assigns positive rewards for correct identification of both normal and anomalous instances, while imposing negative rewards for misclassification. The MSR reward optimizer M aims to optimize the final reward received by A.
- *Reply Memory (R):* The reply memory R stores a set of interactive information at each step t, including the next state s_{t+1}, the current state s_t, the current action a_t and the final reward r_t^*. The random sampled mini-batch from R is adopted to train A.

By making this transformation, the reward function drive A to automatically and interactively learn known anomalies from $\mathcal{D}^a$ and learn unknown anomalies in $\mathcal{D}^u$. The workflow of the DRL-based anomaly detection is as follows:

1) At each time step t, A receives s_t sampled by g and executes an action a_t to maximize the cumulative reward it can obtain. The reward is elaborated in steps 3)–5).
2) Within E, g selects the next state s_{t+1} to A, contingent upon a_t that A has executed on s_t. To effectively leverage both $\mathcal{D}^a$ and $\mathcal{D}^u$, g is designed to return as many possible unlabeled anomalies while at the same time equivalently presenting labeled anomalies to A.
3) f generates an unsupervised reward r_t^i that encourages A to identify unlabeled anomalies within $\mathcal{D}^u$.
4) h confers a positive reward r_t^e if it detects anomalies from $\mathcal{D}^a$ or identifies anomalies/normals correctly in $\mathcal{D}^u$. This mechanism enables A to learn known anomalies while enhancing its ability to detect unknown anomalies.
5) The combined reward r_t, calculated as the aggregate of intrinsic and extrinsic rewards, undergoes optimization. Subsequently, A receives the optimized reward r_t^*.

The agent engages in iterative training, adhering to steps (1)–(5) throughout multiple episodes with each episode comprising a predefined number of states. The agent is trained to optimize cumulative rewards, enabling it to adaptively identify known/unknown anomalies.

IV. PROPOSED WADE

A. Dueling DQN-Based Anomaly Detection Agent A

The anomaly detection agent A aims to learn an optimal anomaly detection-oriented action-value function. Following [74], the action-value function is approximated as:

$$Q^*(s, a) = \max_{\pi} \mathbb{E}\left[R_t = \sum_{t'=t}^{T} \gamma^{t'-t} r_{t'} | s_t = s, a_t = a, \pi\right], \tag{1}$$

which is the maximum expected return when A takes the action a_t for each state s_t following the policy π, with the return defined as the sum of rewards r_t discounted by a factor γ at each time step t.

Given the varying importance of different states in anomaly detection, learning discriminative features of anomalies is particularly crucial in class-imbalanced scenarios. Dueling DQN [19] can effectively assess the contribution of different actions under anomalous states, thereby strengthening the learning of critical anomaly features. Thus, this study adopts Dueling DQN to learn $Q^*(s, a)$ and maximize the expected return for anomaly detection. Dueling DQN employs a deep neural network where the lower layers are convolutional (consistent with the original DQN [74]), followed by two streams of fully-connected layers subsequently. Particularly, it decomposes Q-value function into state and advantage functions, accurately capturing the associations between states and actions under anomaly conditions, with the Q-value function defined as:

$$Q(s,a;\theta,\alpha,\beta) = V(s;\theta,\beta) + \left(A(s,a;\theta,\alpha) - \frac{1}{|\mathcal{A}|}\sum_{a'} A(s,a';\theta,\alpha) \right), \tag{2}$$

where $V(s;\theta,\beta)$ and $A(s,a;\theta,\alpha)$ respectively represent a scalar and a vector of the two streams, and $|\mathcal{A}|$ denotes dimension of the vector. In addition, θ represents parameters of the convolutional layers, while α and β are parameters of the two streams of fully-connected layers.

The network then learns parameters θ, α and β by iteratively minimizing the following loss:

$$\mathcal{L}_j(\theta_j,\alpha_j,\beta_j) = \mathbb{E}_{(s,a,r,s')}\left[(y_j - Q(s,a;\theta_j,\alpha_j,\beta_j))^2 \right], \tag{3a}$$

$$y_j = r + \gamma \max_{a'} \hat{Q}(s',a';\theta_j^-,\alpha_j^-,\beta_j^-), \tag{3b}$$

where (s,a,r,s') is a random mini-batch sampled from the reply memory. a' is the taken action for s'. Parameters θ_j, α_j and β_j belong to the main Q-network, while θ_j^-, α_j^- and β_j^- denote parameters of the target Q-network.

At each step, the action $a_t \in [a^0, a^1]$ is selected as which maximizing $Q(s_t,a;\theta_j,\alpha_j,\beta_j)$ for current state s_t. When evaluating the test set, data instances (or states) with large $Q^*(s,a;\theta^*,\alpha^*,\beta^*)$ values (i.e., high anomaly scores) are supposed to be anomaly-biased data, where θ^*, α^* and β^* are parameters of the trained Q-network.

B. The Reward Optimization Mechanism

To address the challenges of difficult data annotation and the prevalence of diverse unknown attacks in real-world network environments, we design the reward optimization mechanism comprising VAE-based intrinsic reward function f, external reward function h and MSR reward optimizer M. f encourages A to explore potential anomalies in $\mathcal{D}^u$, while h drives A to interactively learn $\mathcal{D}^a$ to achieve high true positive detection and strengthens the awareness of potential anomalies in $\mathcal{D}^u$. M refines the total reward to optimize the decision-making.

1) VAE-Based Intrinsic Reward Function: VAE [20] learns latent data distribution by encoding inputs as parameters (mean and variance) of the latent space. The reconstruction error, which measures the discrepancy between input samples and their reconstructed outputs, serves as an effective indicator for anomaly detection. Specifically, normal samples typically yield small reconstruction errors, while anomalous instances generally produce significantly larger errors.

Consequently, to discover potential anomalies in unlabeled data, the reconstruction error (or loss) of VAE is utilized as the foundation for f. Specifically, we train the VAE model using $\mathcal{D}^u$ with the follow loss function:

$$\mathcal{L}_{\text{vae}} = MSE(x_{\text{recon}}, x) - \frac{1}{2}(1 + \log\sigma^2 - \mu^2 - \sigma^2), \tag{4}$$

where μ and σ^2 denote the mean and variance of the samples, x_{recon} represents the reconstructed output. Then, the intrinsic reward r_t^i for each state s_t is calculated using the reconstruction error as follows:

$$r_t^i = f(s_t) = MSE(s_{t\_\text{recon}}, s_t). \tag{5}$$

Particularly, r_t^i is scaled into the range [0,1] through activation function (Sigmoid) in the decoder network, which ensures a balanced relationship between the reward values of f and h. Thus, the agent will receive large r_t^i when s_t is supposed as potential anomaly, regardless of the specific action a_t taken.

2) External Reward Function: h comprehensively learns potential unlabeled anomalies and labeled anomalies from $\mathcal{D}^u$ and $\mathcal{D}^a$ respectively. For each state s_t, h produces corresponding reward r_t^e based on the taken action (a^0 or a^1) and the intrinsic reward r_t^i from f, which is defined as:

$$r_t^e = h(s_t, a_t, r_t^i) = \begin{cases} 1, & \text{if } a_t = a^1 \text{ and } s_t \in \mathcal{D}^a \\ -1, & \text{if } a_t = a^0 \text{ and } s_t \in \mathcal{D}^a \\ l_t - l_t', & \text{if } a_t = a^1 \text{ and } s_t \in \mathcal{D}^u \\ l_t' - l_t, & \text{if } a_t = a^0 \text{ and } s_t \in \mathcal{D}^u \end{cases}, \tag{6}$$

where $l_t = r_t^i$ and $l_t' = 1 - r_t^i$ respectively represents anomaly and normality degree of the current data instance. For $\mathcal{D}^a$, positive and negative rewards (1 and -1) are assigned when the agent correctly or incorrectly identifies anomalies. For $\mathcal{D}^u$, which contains both unknown normal and abnormal data, the difference between the normality and anomaly degree (l_t' and l_t) is utilized to compute the rewards associated with the taken action a_t. By doing this, the model receives positive rewards through potential correctly identification of both normal and abnormal instances in $\mathcal{D}^u$.

Overall, h maximizes rewards by encouraging the agent to effectively exploit in $\mathcal{D}^a$ and enhance the exploration in $\mathcal{D}^u$ based on f. Compared to the strategy described in [13], which maintains neutrality (provides no reward) in $\mathcal{D}^u$, our design performs better (as discussed in Section V-D).

3) MSR Reward Optimizer: To fully learn both $\mathcal{D}^a$ and $\mathcal{D}^u$, the total reward is designed as $r_t = r_t^i + r_t^e$. However, [21] reveals a phenomenon in DRL that the reward in last step r_{t-1} significantly deviates from the current reward r_t defined as reward saltation. It is categorized as positive saltation or negative saltation when current reward is better or worse. Such reward saltation occurring in DRL-based anomaly detection may result policy collapse and influence detection performance. To address this, we adopt the Magnify Saltatory Reward (MSR) algorithm [21] to dynamically refine the total reward r_t by amplifying it during positive saltation while reducing it during negative saltation.

To determine when to optimize the reward, a parameter ρ is introduced to quantify the magnitude of change between r_t and r_{t-1}, which is defined as follows:

$$\rho = \frac{r_t - r_{t-1}}{\min(|r_t + \sigma|, |r_{t-1} + \sigma|)}, \tag{7}$$

where σ is a number that approaches zero. In addition, to ensure an appropriate magnification of r_t, a monotonically increasing bounded function k is designed as follows:

$$k(x) = \arctan\left(x * \frac{\pi}{2} * \frac{1}{\eta}\right), \tag{8a}$$

$$x = \rho + \mathrm{sgn}(r_t - r_{t-1}), \tag{8b}$$

where η is a certain number as the threshold. When $x > \eta$ or $x < -\eta$, the current reward r_t is optimized as:

$$r_t^* = r_t + (k(x) - \lambda) * |r_t| . \tag{9}$$

C. Anomaly Biased Sampling Function g

Considering how to select s_{t+1}, the anomaly-biased sampling function in [13] is employed as follows:

$$s_{t+1} = g_a(\mathcal{D}^a) \sim U(\mathcal{D}^a), \tag{10a}$$

$$s_{t+1} = g_u(s_t, a_t; \theta^e) = \begin{cases} \arg\min_{s \in \mathcal{S}} d(s_t, s; \theta^e) & \text{if } a_t = a^1 \\ \arg\max_{s \in \mathcal{S}} d(s_t, s; \theta^e) & \text{if } a_t = a^0 \end{cases}, \tag{10b}$$

where $\mathcal{S} \subset \mathcal{D}^u$ is a random subsample, $|\mathcal{S}|$ and the other parameters align with those described in [13]. To empower balanced exploitation and exploration across the entire data, both g_a and g_u have probability 50% being adopted.

This study focuses on the scenarios with a limited number and imbalanced distributions of anomalies. While the adopted anomaly-biased sampling function g essentially drives the selection of data instances surrounding anomalies as s_{t+1}, thereby enhancing the learning of limited anomalies. It also guides A to make optimal decisions, while facilitating continuous learning and strategy optimization through iterative interactions with E. After selecting, s_{t+1} together with s_t, a_t and r_t^* are added into the reply memory R.

D. Algorithms Details

Algorithm 1 outlines the detailed training procedure of WADE. Specially, steps 1-3 initialize the size of replay memory and parameters of the main and target Q-network that are trained with $n\_episodes$ episodes, with each episode including n_steps steps. Steps 7-8 describes the process to select the action a_t, where ϵ-greedy [74] exploration is adopted. In steps 9-10, two sampling functions are used to select s_{t+1}. In steps 11-14, the final reward is computed through reward functions and optimizer. Besides, steps 15-16 store experience in the reply memory and select the mini-batch. The other steps perform the updating process of the Q-network.

After training, the anomaly scores for testing set are calculated as: $score = \arg\max_a Q(s, a; \theta^*, \alpha^*, \beta^*)$ through the trained Q-network based on (2). Specifically, higher anomaly scores generally correspond to greater anomaly degrees.

Algorithm 1: Training Procedure of WADE.

Input: The labeled dataset $\mathcal{D}^a$ and unlabeled dataset $\mathcal{D}^u$
Output: Action-value function $Q(s, a; \theta^*, \alpha^*, \beta^*)$
1: Initialize Q-value function Q with random parameter θ, α and β
2: Initialize target Q-value function $\hat{Q}$ with parameter $\theta^- = 0, \alpha^- = 0$ and $\beta^- = 0$
3: Initialize experience replay memory R to capacity C
4: **for** $j = 1$ to $n\_episodes$ **do**
5: Initial the state $s_1 \sim \mathcal{D}^u$
6: **for** $t = 1$ to n_steps **do**
7: With probability ϵ select a_t from $\{a^0, a^1\}$
8: Otherwise select $a_t = \arg\max_a Q(s_t, a; \theta, \alpha, \beta)$
9: With probability p select s_{t+1} from $\mathcal{D}^a$ using g_a based on Eq. (10a)
10: Otherwise select anomaly-biased s_{t+1} from $\mathcal{D}^u$ using g_u based on Eq. (10b)
11: Compute the intrinsic reward r_t^i by VAE-based intrinsic reward function f based on Eq. (5)
12: Compute the external reward r_t^e by designed external reward function h based on Eq. (6)
13: The combined reward $r_t = r_t^i + r_t^e$
14: Compute the optimized reward r_t^* of r_t by MSR reward optimizer based on Eq. (9)
15: Store experience $(s_t, a_t, r_t^*, s_{t+1})$ in R
16: Randomly sample mini-batch $(s_l, a_l, r_l^*, s_{l+1})$ from R with batch _ size l, received by the agent A
17: Set $y_l = r_l$ if episode terminates at step $l+1$
18: Else set $y_l = r_l + \gamma \max_{a'} \hat{Q}(s_{l+1}, a'; \theta^-, \alpha^-, \beta^-)$
19: Perform gradient descent on $(y_l - Q(s_l, a_l; \theta, \alpha, \beta))^2$ with respect to the network parameter θ, α and β
20: Update $\hat{Q} = Q$ every K steps
21: **end for**
22: **end for**

V. Experiment

A. Datasets and Metrics

1) Datasets: We adopt UNSW-NB15 and NSL-KDD (in Section II-A) to evaluate WADE. Specifically, UNSW-NB15 comprises one normal class and nine anomaly classes. To achieve balanced class distribution, we exclude two rare classes (shellcode and worms) and random downsampling [73] the remaining seven anomaly classes to under 3,000 instances each. For NSL-KDD (groups 39 attack types into four categories), we retain all original anomalies without downsampling in it, simulating a scenario with imbalanced numbers of various anomaly types.

Following Section II-B1, we preprocess the datasets using one-hot encoding and Min-Max normalization. Table VI summarizes the processed datasets, which are divided into training and testing sets with the ratio 4:1 and serve as the foundation for different experimental scenarios.

2) Metrics: AUC-ROC and AUC-PR are adopted as evaluation metrics, with values ranging from [0,1] where higher scores indicate better performance. The reported values represent averages from 10 independent runs, with standard deviations calculated to assess model stability. Following [11], the average ranking is also calculated across all datasets.

TABLE VI
DATASETS SUMMARY. *D* REPRESENTS DATA DIMENSIONALITY, *N*-SIZE IS THE SIZE OF NORMAL DATA, *A*-CLASS CONTAINS ANOMALY CLASSES OF THE DATASET, AND *A*-SIZE IS THE SIZE OF EACH ANOMALY CLASS.

Dataset	Size	*D*	*N*-size	*A*-class	*A*-size	Ratio
UNSW-NB15	113006	196	93000	Analysis	2677	17.7%
				Backdoor	2329	
				DoS	3000	
				Exploits	3000	
				Fuzzers	3000	
				Generic	3000	
				Reconnaissance	3000	
NSL-KDD	148517	122	77054	DoS	53563	48.1%
				R2l	3569	
				Probe	14079	
				U2r	252	

B. Baseline Methods and Parameter Settings

1) Baseline Methods: Considering the IoT scenarios with limited labeled anomalies and abundant unlabeled data, we adopt four weakly supervised methods (DPLAN [13], DevNet [73], FEAWAD [18] and PReNet [17]) in Section II-E for comparison. These baselines employ diverse detection principles, offering comprehensive performance benchmarks.

2) Parameter Settings: All baselines follow their original recommended configurations for optimal performance. Then a single hidden layer with 20 units and ReLU activation is applied in the network architecture of all methods. For WADE, we train VAE model with 15 episodes ($batchsize = 128$) and set the MSR threshold $\eta = 2$. Other parameter settings are consistent with DPLAN, including episodes $n\_episodes = 20$, each episode steps $N^e = 2000$, warming-up steps $N^w = 10000$, target network updated steps $N^t = 30000$ [13] and default settings of the original DQN [74].

C. Anomaly Detection Performance

1) Known Anomalies From One Class: In this scenario, $\mathcal{D}^a$ contains only one specific class of anomalies with a fixed number $n = 60$. Contamination rate $c = 0.02$ represents the ratio of abnormal data to total data in $\mathcal{D}^u$, where almost all classes of anomalies are controlled to a certain quantity. After processing, two basic datasets are divided into 11 sub-datasets (each named by specific anomaly class). Here, $\mathcal{D}^a$ account for 0.05%-0.07% of the training sets for different sub-datasets.

Table VII presents the experimental results across all sub-datasets. Overall, WADE achieves the best AUC-PR on 7 sub-datasets and AUC-ROC on 6 sub-datasets. Compared to baselines, WADE averagely obtains AUC-PR improvement of 3.5% (DPLAN), 19.9% (FEAWAD), 12.2% (DevNet), 13.3% (PReNet) and AUC-ROC improvement of 1.4% (DPLAN), 17.6% (FEAWAD), 4.7% (DevNet), 2.8% (PReNet). Across different datasets, WADE performs well on *UNSW-NB15*. While for *NSL-KDD*, it achieves better results on *R2l* and *U2r* (containing fewer anomalies) but shows slightly reduced performance on *DoS* and *Probe* (containing more anomalies). This discrepancy likely stems from the imbalanced distribution of anomaly samples across different classes, further confirming WADE's strength in scenarios with severely limited anomalies. Moreover, WADE exhibits remarkable stability across all sub-datasets, as indicated by its lower standard deviation values.

2) Larger Amount and Multiple Classes in Known Anomalies: This section examines how the quantity and diversity of labeled anomalies influence detection performance, where the anomaly amount and classes in $\mathcal{D}^a$ are increased. Considering the diverse anomaly classes in UNSW-NB15, we extend the seven sub-datasets from it by incrementally adding 1-6 anomaly classes to $\mathcal{D}^a$, which correspondingly increases the known anomaly classes in testing set. Then six experimental groups are conducted, each randomly selecting 1-6 anomaly classes from unknown anomalies (fixed at 60 samples per class). For comprehensive comparison, we introduces WADE* which maintains the original anomaly classes while increasing sample counts to match the total anomaly quantity. Comparative results are presented in Fig. 3–4.

Experimental results demonstrate significant performance improvements for WADE as the increasing of anomaly classes and amount. Specially, AUC-PR increases remarkably from 0.544-0.707 up to 0.741-0.775 and AUC-ROC increases from 0.843-0.906 up to 0.928-0.935. Meanwhile, the performance of DPLAN generates more fluctuations indicating that WADE is better at leveraging known supervision information. Furthermore, WADE consistently outperforms WADE* with the same number of anomalies, demonstrating that expanding the diversity of anomaly classes enhances the coverage of known anomalies and yields more supervision information.

3) Increasing Anomaly Pollution: To evaluate model robustness in IoT environments where training data may contain unknown anomalies, we systematically vary the contamination rate c of $\mathcal{D}^u$ across $\{0.02, 0.04, 0.06, 0.08, 0.1\}$. Figs. 5/6 present the AUC-PR/ROC performance under these conditions. Overall, all methods exhibit clear performance degradation on most datasets as c increases. While WADE maintains superior performance with a less pronounced decline compared to baselines under varying contamination rates. Notably, WADE shows relatively stable performance on Exploits, Generic, DoS, Probe and U2r. This stability stems from its enhanced ability to learn potential anomalies from $\mathcal{D}^u$, confirming that learning known and unknown anomalies can appropriately tolerate increasing noise levels. Moreover, compared to DPLAN (another DRL method), WADE exhibits smaller performance variation, validating our architectural improvements substantially enhance model robustness.

D. Ablation Study

This section presents a systematic ablation study to evaluate the impact of key components on detection performance of WADE. Four variants are examined by respectively removing: 1) intrinsic reward (*DME*), 2) MSR algorithm (*DVE*), 3) external reward (*DVM*) and 4) Dueling DQN (*VME*). The results are summarized in Table VIII.

Experiments on DME: DME variant excludes intrinsic reward r_t^i when computing the total reward r_t, resulting in the decreases of 3.7% in AUC-ROC and 1.8% in AUC-PR. This

TABLE VII
DETECTION PERFORMANCE (AUC-ROC/AUC-PR ± STANDARD DEVIATION) OF WADE AND BASELINE METHODS

	AUC-ROC Performance					AUC-PR Performance				
Dataset	WADE	DPLAN	FEAWAD	DevNet	PReNet	WADE	DPLAN	FEAWAD	DevNet	PReNet
Analysis	**0.889**±0.025	0.870±0.020	0.846±0.019	0.875±0.019	0.862±0.023	**0.687**±0.027	0.673±0.025	0.571±0.019	0.636±0.017	0.625±0.013
Backdoor	**0.867**±0.016	0.861±0.014	0.825±0.029	0.861±0.006	0.826±0.001	**0.654**±0.024	0.637±0.023	0.601±0.017	0.636±0.005	0.606±0.002
DoS	**0.906**±0.013	0.901±0.012	0.878±0.019	0.887±0.005	0.835±0.002	**0.707**±0.008	0.695±0.017	0.612±0.025	0.627±0.004	0.596±0.004
Exploits	**0.865**±0.016	0.860±0.019	0.785±0.050	0.801±0.019	0.786±0.004	**0.632**±0.033	0.619±0.033	0.530±0.024	0.358±0.038	0.325±0.007
Fuzzers	0.843±0.018	**0.859**±0.013	0.811±0.013	0.818±0.008	0.835±0.005	0.517±0.035	**0.531**±0.031	0.419±0.020	0.389±0.020	0.393±0.014
Generic	**0.843**±0.034	0.832±0.024	0.704±0.041	0.761±0.020	0.752±0.007	**0.677**±0.013	0.651±0.021	0.541±0.036	0.573±0.018	0.545±0.016
Reconnaissance	0.851±0.012	0.846±0.012	0.756±0.039	0.864±0.003	**0.871**±0.002	0.544±0.025	0.502±0.024	0.501±0.026	0.515±0.011	**0.575**±0.005
DoS	0.920±0.025	0.877±0.032	0.926±0.011	0.927±0.010	**0.954**±0.019	0.944±0.014	0.905±0.022	0.954±0.005	0.953±0.004	**0.964**±0.009
R2l	**0.826**±0.027	0.781±0.066	0.407±0.048	0.717±0.028	0.823±0.005	**0.762**±0.049	0.728±0.058	0.487±0.048	0.675±0.012	0.644±0.016
Probe	0.926±0.025	0.898±0.026	0.690±0.050	0.962±0.004	**0.964**±0.002	0.920±0.031	0.881±0.026	0.793±0.031	0.957±0.006	**0.958**±0.006
U2r	0.810±0.067	**0.839**±0.056	0.491±0.047	0.651±0.048	0.771±0.011	**0.832**±0.061	0.792±0.074	0.560±0.030	0.697±0.012	0.726±0.008
Average	**0.868**±0.025	0.856±0.027	0.738±0.033	0.829±0.012	0.844±0.007	**0.716**±0.029	0.692±0.032	0.597±0.026	0.638±0.013	0.632±0.009
Average.Rank	**1.8181**	2.6363	4.8181	2.8181	2.9090	**1.6363**	2.5454	4.2727	3.2727	3.1818

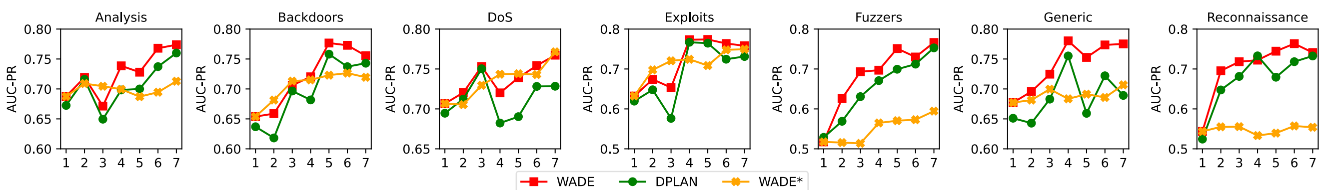

Fig. 3. AUC-PR performance of WADE, DPLAN and WADE* w.r.t. the number of known anomaly classes.

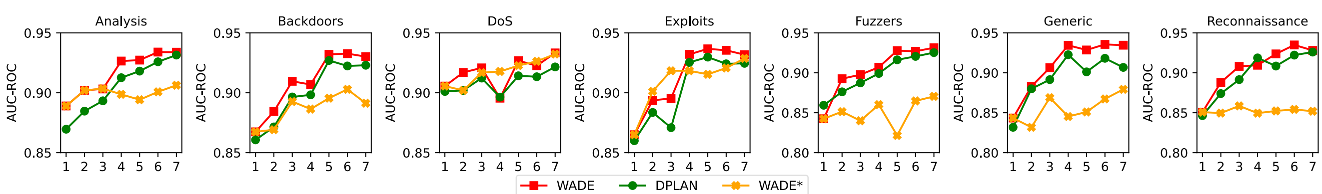

Fig. 4. AUC-ROC performance of WADE, DPLAN and WADE* w.r.t. the number of known anomaly classes.

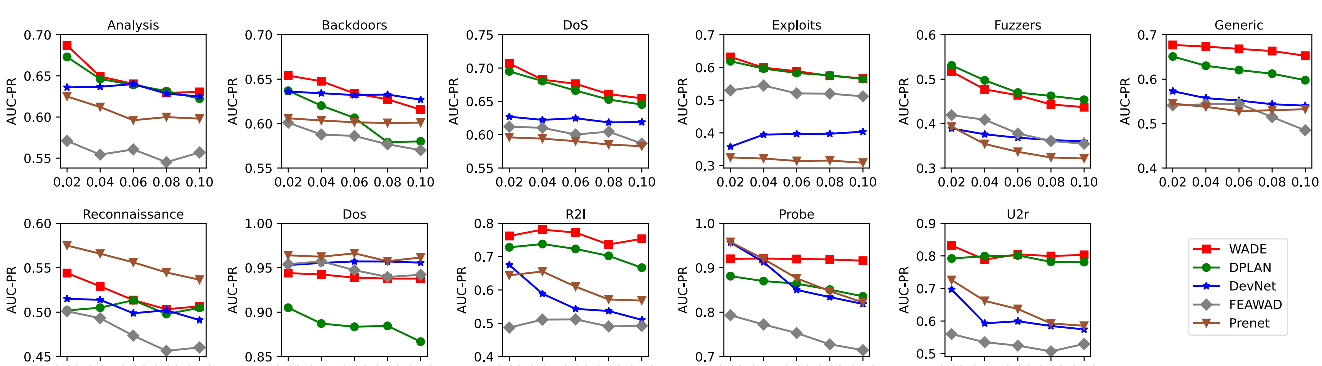

Fig. 5. AUC-PR performance of WADE and its competing methods w.r.t. different contamination rates.

discrepancy stems from the distinct learning focuses of r_t^i and r_t^e. The former primarily learns from unlabeled data (mainly normal samples), while the latter specializes in learning labeled anomalies. AUC-ROC reflects the overall discriminative ability for positive/negative samples, removing r_t^i (which focuses on learning normals) causes significant metric decline. In contrast, the learning of labeled anomalies by r_t^e is less affected, resulting in a relatively smaller decline in AUC-PR, which specifically evaluates anomaly detection performance. Building upon DME, we further develop DME* by eliminating intrinsic rewards from the extrinsic reward function design. Experimental results show that DME* achieves inferior performance (AUC-ROC: 0.825%,

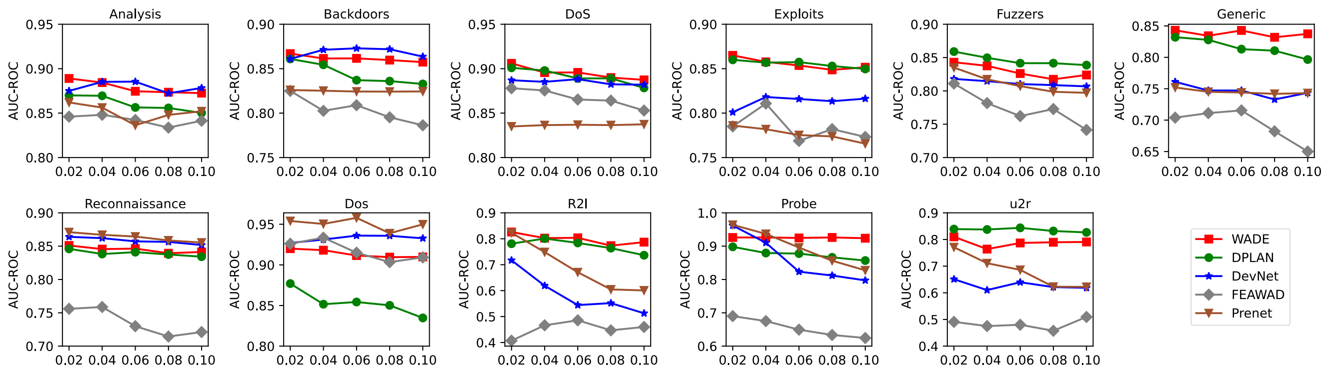

Fig. 6. AUC-ROC performance of WADE and its competing methods w.r.t. different contamination rates.

TABLE VIII
ABLATION EXPERIMENT RESULTS

	AUC-ROC Performance					AUC-PR Performance				
Dataset	**WADE**	**DME**	**DVE**	**DVM**	**VME**	**WADE**	**DME**	**DVE**	**DVM**	**VME**
Analysis	0.889±0.025	0.889±0.021	0.889±0.017	**0.893**±0.013	0.866±0.026	**0.687**±0.027	0.684±0.033	0.685±0.016	0.683±0.019	0.633±0.047
Backdoor	0.867±0.016	**0.869**±0.011	0.868±0.012	0.865±0.011	0.849±0.016	0.654±0.024	0.650±0.029	**0.659**±0.021	0.649±0.014	0.575±0.039
DoS	0.906±0.013	0.905±0.015	0.906±0.014	**0.910**±0.008	0.877±0.018	0.707±0.008	0.705±0.028	**0.714**±0.022	0.706±0.018	0.632±0.049
Exploits	**0.865**±0.016	0.840±0.030	0.865±0.015	0.837±0.030	0.843±0.017	**0.632**±0.033	0.615±0.043	0.620±0.026	0.615±0.034	0.564±0.037
Fuzzers	**0.843**±0.018	0.827±0.016	0.836±0.021	0.819±0.020	0.831±0.023	**0.517**±0.035	0.496±0.016	0.487±0.034	0.494±0.021	0.476±0.052
Generic	0.843±0.034	0.808±0.028	0.823±0.029	0.790±0.005	**0.851**±0.024	0.677±0.013	0.683±0.023	0.680±0.015	**0.694**±0.011	0.652±0.026
Reconnaissance	**0.851**±0.012	0.845±0.020	0.848±0.014	0.841±0.022	0.844±0.016	**0.544**±0.025	0.543±0.018	0.536±0.030	0.534±0.030	0.515±0.027
DoS	**0.920**±0.025	0.920±0.023	0.912±0.030	0.915±0.031	0.854±0.057	**0.944**±0.014	0.941±0.016	0.940±0.016	0.941±0.020	0.895±0.040
R2l	**0.826**±0.027	0.725±0.073	0.738±0.084	0.681±0.075	0.819±0.082	0.762±0.049	0.748±0.041	0.713±0.066	0.760±0.052	**0.796**±0.072
Probe	**0.926**±0.025	0.870±0.131	0.905±0.060	0.839±0.174	0.896±0.044	**0.920**±0.031	0.891±0.066	0.908±0.053	0.883±0.085	0.887±0.038
U2r	**0.810**±0.067	0.694±0.122	0.649±0.075	0.689±0.083	0.741±0.131	**0.832**±0.061	0.773±0.085	0.720±0.071	0.762±0.103	0.736±0.109
Average	**0.868**±0.025	0.836±0.045	0.840±0.033	0.825±0.043	0.843±0.041	**0.716**±0.029	0.703±0.036	0.697±0.034	0.702±0.032	0.669±0.049

TABLE IX
TRAINING TIME COMPARISON OF DIFFERENT METHODS

Methods	**Training time (2,000 Steps)**
WADE	82.33s
WADE w/o MSR	80.86s
DPLAN	81.59s
FEAWAD	91.23s
DevNet	77.34s
PReNet	80.17s

AUC-PR: 0.673%) compared to DME, providing additional empirical evidence for the beneficial role of intrinsic rewards.

Experiments on DVE: DVE variant computes the combined reward through direct summation of intrinsic and external rewards (without MSR optimization), resulting in average performance degradation of 3.2% in AUC-ROC and 2.6% in AUC-PR. Notably, on highly imbalanced datasets like U2r and R2l where anomalies are extremely scarce, removing the MSR module leads to pronounced degradation in detection performance. These results demonstrate that the MSR reward optimization module effectively stabilizes training under extreme anomaly sparsity and significantly improves the model's ability to learn rare anomalies. Furthermore, comparative training time experiments (in Table IX) confirm that the MSR module introduces negligible computational overhead.

Experiments on DVM: DVM variant incorporates the external reward function mentioned in [13]. In this design, negative reward of -1 is given when $a_t = a^1$ and $s_t \in \mathcal{D}^u$, while no reward is given when $a_t = a^0$ and $s_t \in \mathcal{D}^u$. DVM obtains average reductions of 5.0% in AUC-ROC and 2.0% in AUC-PR, validating the efficacy of the designed external reward function h. Specifically, [13] treats unlabeled data as exclusively normal but penalizes anomalies inherently degrading discrimination performance. In contrast, WADE assigns positive rewards for correct identification of both normal and anomalous instances within $\mathcal{D}^u$. This optimization framework maximizes cumulative rewards and enhances discriminative learning, leading to superior performance.

Experiments on VME: VME variant substitutes the Dueling DQN with a standard DQN architecture, resulting in performance declines of 3.0% in AUC-ROC and 6.5% in AUC-PR. This significant decline, particularly in AUC-PR, confirms that the separate estimation of state and advantage values in the Dueling DQN contributes to anomaly detection tasks. Notably, the standard DQN shows marginally better performance on the Generic and R2l, likely due to the highly localized nature of anomaly patterns in these data features that may not benefit from the separation of state and advantage functions provided by the Dueling DQN.

E. Convergence Analysis

Considering that reward curves effectively reflect the convergence behavior of DRL, we plot the training reward curves across all datasets (in Fig. 7). The results show that WADE

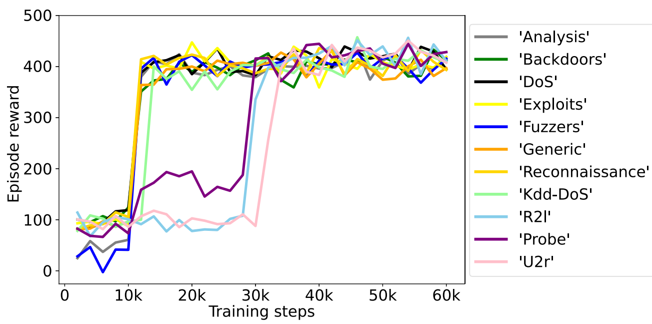

Fig. 7. Episode reward w.r.t. training steps.

achieves early reward stabilization across all datasets, typically converging within 30,000 steps, which supports our adoption of the 30,000-step training setting. Additional, the observed slower convergence of WADE on U2r, Probe and R2l datasets can be attributed to their constrained sample sizes and severe class imbalance, both of which adversely affect strategy optimization and convergence stability.

F. Training Time Analysis

Table IX presents the training time (2,000 steps) of the baseline methods. Compared to DL-based methods, WADE incurs lower training overhead than FEAWAD but slightly higher than DevNet and PReNet. This moderate increase in training time primarily stems from: 1) the complex model architecture incorporating additional VAE training, and 2) the dual-network parameter updates (Q-network and target Q-network) in the agent. Notably, WADE achieves comparable training efficiency to DRL-based DPLAN, demonstrating that WADE increases no computational complexity.

Considerding model deployment, WADE exhibits inherent real-time anomaly detection capabilities. It integrates streaming data into the replay memory, enabling dynamic parameter updates through randomized batch sampling without requiring complete model retraining. This efficient update mechanism provides robust adaptability to data streams with experiments demonstrating its computational efficiency (2000 steps in 82 s), indicating strong potential for real-time deployment.

VI. Conclusion

This paper proposes a DRL-based weakly supervised traffic anomaly detection method (WADE) for IoT networks. WADE tackles the challenge of detecting unknown anomalies with limited labeled anomalies (containing few known classes) and large-scale unlabeled data (with both normal and anomalous instances). Specifically, WADE adopts Dueling DQN to design the anomaly detection-oriented agent. Then, to enhance the environment-agent interaction and unlabeled data exploration, WADE introduces a reward optimization mechanism and an anomaly biased sampling function. The reward mechanism comprises VAE-based intrinsic reward function, external reward function and MSR reward optimizer. Experimental results on real-world traffic datasets demonstrate the superior performance of WADE in detecting both known and unknown anomalies compared to four semi-/weakly supervised methods.

This study significantly enhances IoT network security by enabling timely detection of attacks within large-scale network traffic. Future research will focus on advancing zero-shot and multi-classification anomaly detection approaches, through the integration of LLMs, optimized data preprocessing and representation learning methods.

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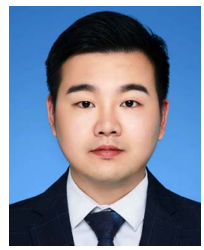

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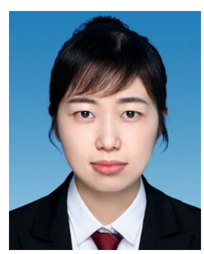

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